Carleton University

Department of Systems and Computer Engineering

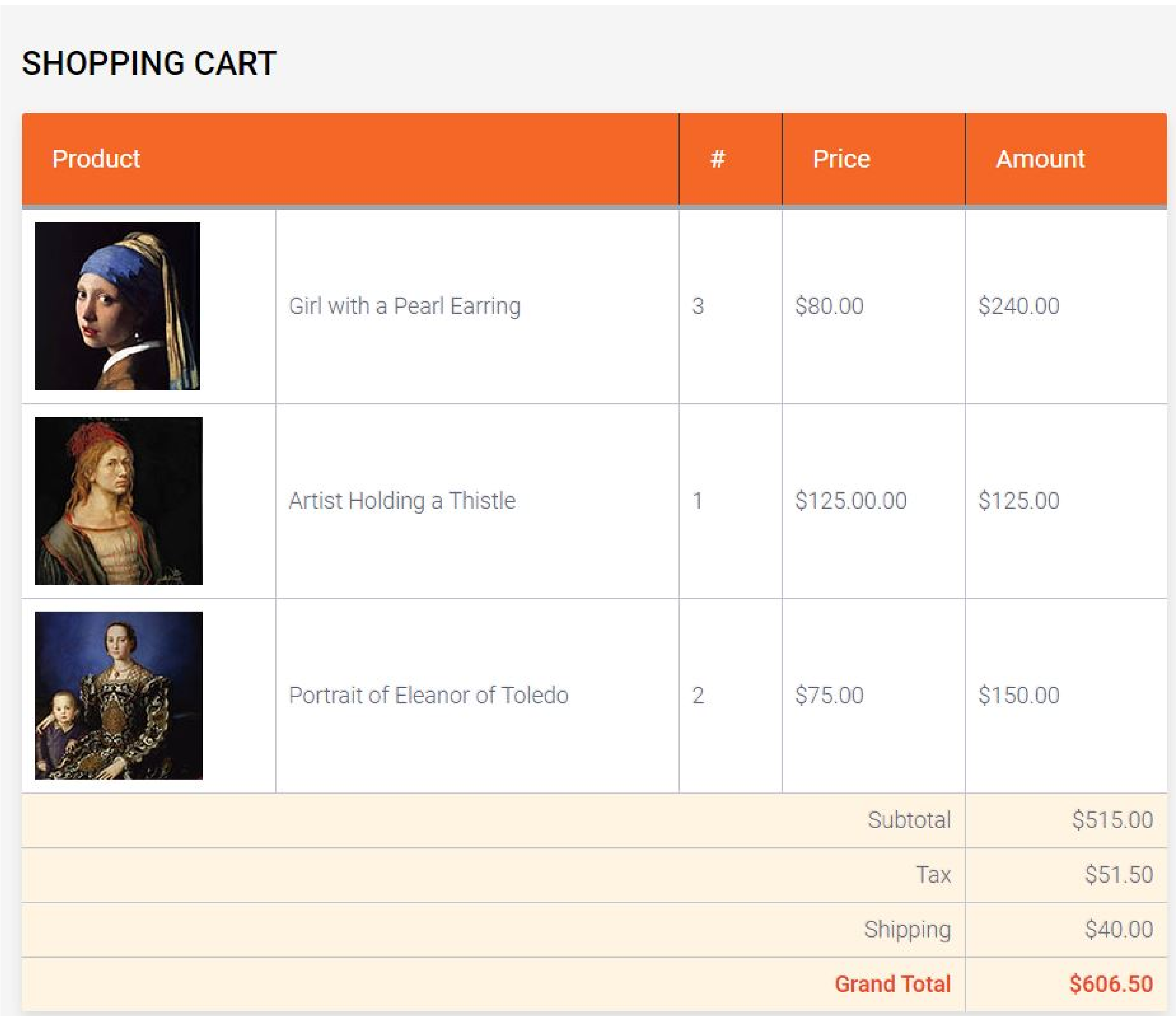
SYSC 4504 *Fundamentals of Web Development* *Winter 2026*

Lab Handout

Lab 3: JavaScript

Due: At the end of the assigned lab session

This lab focuses on JavaScript. You are asked work with JavaScript objects and to populate a web page programmatically, rather than hard-coding that information with HTML markup. You can download a set of files (as ZIP archive) to get you started with the project from `Brightspace`, containing an HTML document called `ArtStore.html` and three subdirectories `images`, `css`, and `js` (with the obvious meaning: `images` contains images used in this project, `css` any styling, and `js` eventually the JavaScript code you write). When you open `ArtStore.html` in a browser (here Microsoft Edge), it looks as follows:



SHOPPING CART

Product		#	Price	Amount
	Girl with a Pearl Earring	3	$80.00	$240.00
	Artist Holding a Thistle	1	$125.00.00	$125.00
	Portrait of Eleanor of Toledo	2	$75.00	$150.00
			Subtotal	$515.00
			Tax	$51.50
			Shipping	$40.00
			Grand Total	**$606.50**

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The task will be to replace the markup for the three images with a JavaScript loop, as described below.

1. Examine the data file `data.js`. It contains four arrays that you are going to use to programmatically generate the data rows (and replace the hard-coded markup in the supplied HTML file).
2. In the file `functions.js` create a function called `calculateTotal()` that is passed a quantity and price and returns their product (i.e., multiple the two parameter values and return the result).
3. Within `functions.js`, create a function called `outputCartRow()` that has the following signature:

    ```
    function outputCartRow(file, title, quantity, price, total)
    ```

    Implement the body of that function. It should use `document.write()` calls to display a row of the table using the passed data. Use the `toFixed()` method of the number variables to display two decimal places.
4. Replace the three cart rows in the original markup with a JavaScript loop that repeatedly calls this `outputCartRow()` function. Put this loop in a file called `ArtStore.js` in the `js` subdirectory. Include the appropriate `<script>` tag to reference this `ArtStore.js` within the `<tbody>` element.
5. Calculate the subtotal, tax, shipping, and grand total using JavaScript. Replace the hard-coded values in the markup with your JavaScript calculations. Use 10% as the tax amount. The shipping should be $40 unless the subtotal is above $1000, in which case it will be $0 (i.e., free shipping for good customers).

Test the page in the browser. Verify that the calculations work appropriately by changing the values in the `data.js` file.

Submission: create a ZIP archive that contains your complete final solution, including the modified `ArtStore.html` and all subdirectories. Upload this ZIP file as your lab submission. The TA should be able to simply double-click on your `ArtStore.html` and see an output similar to the figure above (after extracting everything from your submitted ZIP file). DO NOT USE RAR or any other archive format, and DO NOT submit the files individually.